

Hamza Farach

Curriculum Vitae

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🔗 github.com/hamzafarach

Education

- 2025–Present **Master of Research (Computer Science)**, *Keio University*, Tokyo, Japan.
GPA: 3.83/4.0 | Dual Degree Program with Ecole Centrale de Nantes
Main courses: Computer Vision, NLP, Statistical Mechanics, Statistics, Advanced Signal Processing.
- 2023–Present **Master of Engineering (AI for Image and Signal Processing)**, *Ecole Centrale de Nantes*, Nantes, France.
GPA: 3.99/4.0 | **Admission:** Top 3.1% in national entrance exams.
Main courses: Mathematical Optimization, Probability Theory & Statistics, Inverse Problems, Deep Learning, Image Processing.
- 2020–2023 **Preparatory Classes (MPSI-MP)**, *Lycée Louis Le Grand*, Paris, France.
Highly selective two-year curriculum (ranked top 1% nationally).
Main courses: Mathematics, Physics, Computer Science.

Publications & Patents

- 2026 **Hamza Farach**, Mondher Bouazizi, Tomoaki Ohtsuki – "Occlusion-Robust LiDAR Human Segmentation via Geometry-Consistent Occlusion Synthesis" - *Submitted to IEEE Sensors*, 2026.
- 2026 **Hamza Farach**, Zoubida Ameer, Frederic Lefebvre, Philippe De Lagrange, Miloš Radosavljević – "Film Grain Needs Attention" - *Accepted at SPIE*, Aug. 2026.
- 2025 **Hamza Farach** & InterDigital R&D Team – "2 Invention Patents regarding Deep Learning for Film Grain Analysis" - *Filed by InterDigital R&D France*, 2025.

Research Experience

- Oct. 2025 – **Graduate Researcher**, *Keio University*, Tokyo, Japan.
Present **Supervisor:** Prof. Tomoaki Ohtsuki
Conducting research in computer vision, focusing on human perception under severe occlusion for 3D LiDAR depth data in the context of Human Activity Recognition (HAR).
- Apr. 2025 – **Research Intern**, *InterDigital R&D (Video Lab)*, Rennes, France.
Sep. 2025 **Supervisors:** Zoubida Ameer, Frederic Lefebvre
Developed FGA-ANN, an attention-based deep learning model achieving state-of-the-art performance for normative film grain analysis. Designed the first deep learning-based similarity metric tailored to film grain characteristics. *Outputs: 1 paper accepted at SPIE 2026, 2 filed patents.*

Computer & Language Skills

- Programming Python (PyTorch), MATLAB, C++, SQL
Languages English (Fluent), French (Native), Arabic (Fluent), Spanish (Intermediate)

Referees

Prof. Tomoaki Ohtsuki
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